



**FACULTY OF ECONOMIC AND MANAGEMENT SCIENCES
DEPARTMENT OF ECONOMICS**

ECONOMICS 114 A3

DATE:	15 June 2023
DURATION:	3 hours
TOTAL OF MARKS:	100

**SECTION C
FEEDBACK**

QUESTION 26

The majority of students answered this question correctly (-2.35%). The biggest problems were not showing negative value and/or not properly calculating the growth rate as a percentage.

QUESTION 27

27.1. Students needed to give a logical explanation for why an increase in AVERAGE income would only be temporary according to Malthus's theory. This involves a few steps - increase in average income leads to population increase and this results in decreased average incomes due to diminishing marginal productivity. Incomes remained at subsistence levels for very long periods of time. Students who wrote nonsense, e.g. that population remains constant or declines as a result of technological improvements, were penalized accordingly.

27.2. Many students simply wrote down factors that can lead to economic growth, e.g. technological innovation, free markets, trade, etc., but these factors are not enough to explain an ESCAPE from the Malthusian trap. Something more is needed.

QUESTION 28

Q28.1 and 28.2 These questions were generally well answered. Most students were able to correctly calculate maximum income as R150, and the opportunity cost of an hour of free time to be equivalent to the wage rate.

Q28.3: Some students were unsure how to specify the budget constraint in question 28.3.1 using the new wage rate and the R100 of unearned income. Considering that $c = w(24 - t)$, where w = hourly wage rate and t = hours of free time, then w would now be 15, and the R100 would be a constant in the budget constraint (i.e. consumption can be up to R100 even if no hours are worked). For the final part of the question (28.3.2), students needed to argue that free time

would decrease, since the substitution is negative due to free time now being relatively more expensive (opportunity cost has increased).

QUESTION 29

Q29.1: Because the indifference curve is tangent to BL1 at A, the $(-)MRS = \text{slope of BL1 (MRT)}$. This means $MRS = 2$. Students would only receive 1 mark if they indicated MRS as negative.

Q29.2: If $MRT = 2$, then $P_x/P_y = 2$, making $P_y = R2$. The income could be calculated using the quantities at point A and the two prices (R100) or using either of the intercepts of BL1. Some students incorrectly calculated income using both intercepts.

Q29.3: Generally, not answered well. BL2 has a slope of $1/2$, which represents the relative price of X. Many students gave this as the answer: $Y = 1/2X$. But it should be the reverse ($Y = 2X$), since $1/2 = P_x/P_y$, and so $P_y/P_x = 2$. A relative price is not a Rand amount; it tells us how many units of X could have been bought with the amount spent on a unit of Y.

QUESTION 30

Most students did well in this question. The Nash equilibrium of this game is (Don't wash dishes; Don't wash dishes) or (5,5). The reason why this is a social dilemma is that where it might be tempting for one person to not wash dishes, if both Leonard and Sheldon decide not to wash dishes, it is to the detriment of both of them (i.e. the dirty dishes continue piling up). Both Leonard and Sheldon would have been better off if they chose to wash the dishes.

QUESTION 31

Q31.1: The answer here was G, because the distance between the FF (representing the total protein produced) and the x-axis is largest. Note that "preferred" doesn't have to correspond with "feasible".

Q31.2: The answer required here was "somewhere between C and D", and most likely midway (given equal bargaining power). Any point between C and D is Pareto efficient ($MRT = MRS$). However, neither C nor D will be the outcome with equal bargaining power: the outcome C puts Wachowski in the same position as his reservation option (and maximises Keanu's economic rent), and outcome D puts Keanu in the same position as his reservation option (and maximises Wachowski's economic rent). This means that, with some bargaining power, they will both try to be better off than their reservation option, and they will end up somewhere between C and D.

QUESTION 32

$$\begin{aligned}\text{employment rent per hour} &= \text{wage} - \text{reservation wage} - \text{disutility} \\ &= 15 - 5 + 2 - 1 \\ &= R11\end{aligned}$$

$$\begin{aligned}\text{Total employment rent} &= \text{employment rent per hour} \times \text{expected hours lost} \\ &= 11 \times (44 \times 40) \\ &= R 19360\end{aligned}$$

Note: Most students did not recognise that the psychological and social costs decreased the reservation wage.

QUESTION 33

33.1 Most students did well in question 33.1. We know that equilibrium price and quantity are determined where the demand and supply curves intersect. Therefore, we can set demand equal to supply to determine the equilibrium quantity and once we have the value for equilibrium quantity, we can substitute this value in either the demand or supply function to calculate the equilibrium price.

The inverse demand function is given as: $P = 68 - 0.02Q$

The inverse supply function is given as: $P = 28 + 0.005Q$

We set demand equal to supply to solve Q:

$$\begin{aligned}68 - 0.02Q &= 28 + 0.005Q \\ \therefore 0.025Q &= 40 \\ \therefore Q &= 1600\end{aligned}$$

To solve equilibrium price, we can substitute this value for Q (1600) into either the demand or the supply function:

$$\begin{aligned}P &= 68 - 0.02(1600) \\ \text{Therefore, } P &= 36\end{aligned}$$

Therefore, equilibrium quantity (Q) = 1600
Equilibrium Price (P) = 36

33.2 Very few students got this question right. Most students tried to get the answer by trying to do a calculation of the value of the price elasticity of demand, but this was not necessary. If you read the first part of the question carefully, you will find the answer there. "Julian's Chesa-Nyama is one of many businesses in a competitive braai food market...". This means that it is a perfectly competitive market. We know that in a perfectly competitive market, the demand curve is perfectly elastic.

- 33.3 Very few students managed to get this question right. It is important to start off by stating where profit-maximisation takes place, i.e. where $MR = MC$.

Since we are working in a perfectly competitive market, we know that $P = MR$. We calculated P in question 33.1 ($P = 36$), therefore: $MR = 36$.

MC is calculated by taking the first derivative of TC . Note that the first derivative is NOT calculated by dividing by q . You need to differentiate the TC equation in terms of q to get to MC .

$$TC = 40 + 28q + 1/8q^2$$

$$MC = dTC/dq = 28 + 1/4q$$

To determine how many braai meals Julian should sell to maximise his economic profit, we set $MC = MR$. Therefore:

$$28 + 1/4q = 36$$

And now we solve the value of q :

$$1/4q = 8$$

Therefore, $q = 32$

- 33.4 Most students knew the formula to calculate economic profit, i.e. Profit = Total revenue – Total cost
Or: $\pi = TR - TC$

$$\therefore \pi = (P \times q) - (40 + 28q + 1/8q^2)$$

We calculated the value of P in question 33.1 ($P = 36$). In Question 33.3 we calculated the value of q (we know that $q = 32$).

$$\therefore \pi = (36 \times 32) - [40 + 28(32) + (1/8)(32)^2]$$

$$\therefore \pi = 1152 - [40 + 896 + 128]$$

$$\therefore \pi = 1152 - 1064$$

$$\therefore \pi = 88$$

- 33.5 Most students did not get this question right and argued that Julian can increase his profit by raising the price of his meals, which is not correct. The correct answer is No, since Julian is operating in a perfectly competitive market, which makes him a price taker. A perfectly competitive firm does not have any market power. If Julian was to

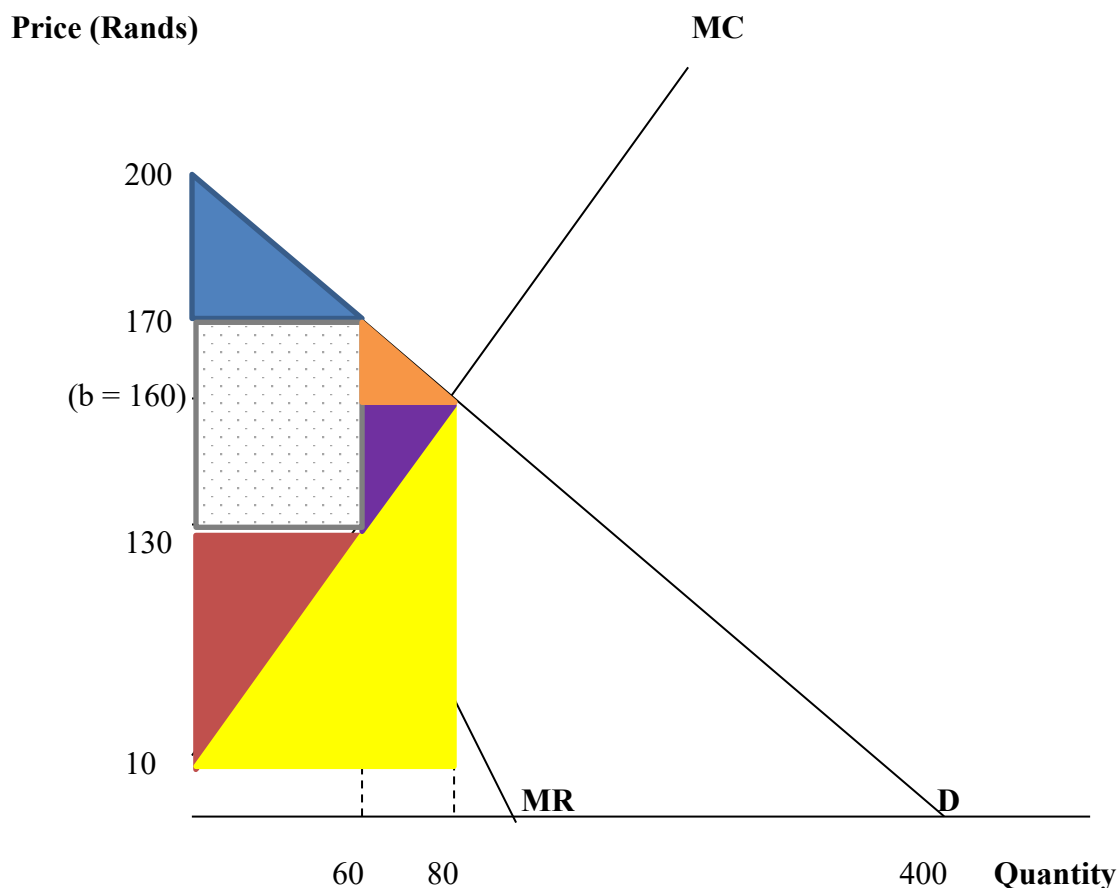
increase the price of his meals, the demand for his meals will fall to zero and he will not have any sales.

QUESTION 34

This question was answered well.

- 34.1 The majority of students correctly stated that there is a shortage in the market due to the price ceiling, but often their calculations were wrong, and thus they did not necessarily show a shortage of 18. Students had to show that quantity demanded is 21 whilst quantity supplied is 3.
- 34.2 The results were mixed, but many did correctly state that consumer surplus will increase.

QUESTION 35



- 35.1 Most students did well in this question.

To calculate the slope of the demand curve, you could either use the total big triangle (starting from point 200 on the y-axis, moving down to the origin – where zero is - and then moving to the right to point 400 on the x-axis). You will then calculate the slope as follows:

$$\frac{\Delta y}{\Delta x} = \frac{(200-0)}{(400-0)} = 0.5 \text{ (which is a negative value, since the demand curve slopes negatively).}$$

If, instead, you used the blue  triangle in the graph above to calculate the slope of the demand curve, you would get:

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{(200-170)}{(60-0)} = 0.5 \text{ (remember that this slope is negative, since the demand curve's slope is negative).}$$

This would then give you the following equation for the demand curve: $P = -0.5Q + 200$ (where 200 is the y-intercept).

You could also write the demand equation as: $Q = -2P + 400$

To calculate the slope of the MC curve, you could either calculate the slope of the little orange

() triangle in the graph above, or calculate the slope using the little yellow () triangle or

calculate the slope using the little purple triangle () .

Let's look at the orange triangle first:

$$\text{Slope (using the orange triangle first)} = \frac{\Delta y}{\Delta x} = \frac{(130-10)}{(60-0)} = 2$$

Therefore, **MC = 2Q + 10 (where 10 is the y-intercept).**

Alternatively, when using the yellow triangle, you would first need to get the value for "b" (which is a point I added on the graph). We can use the equation for the demand curve to get to the value of b:

$$P = -0.5Q + 200$$

When $Q = 80$, $P = 160$ (which is b's value)

$$\text{Slope (using the yellow triangle)} = \frac{\Delta y}{\Delta x} = \frac{(160-10)}{(80-0)} = 1.875$$

This will then give you the following MC equation: **MC = 1.875Q + 10** (where 10 is the y-intercept).

Alternatively, if you use the purple triangle, you will calculate the slope as follows:

$$\text{Slope (using the purple triangle)} = \frac{\Delta y}{\Delta x} = \frac{(160-130)}{(80-60)} = 1.5$$

Therefore, **MC = 1.5Q + 10 (where 10 is the y-intercept).**

You will therefore get different values for the slope of the MC curve, depending on which coordinates you use to do your calculations, which is strictly speaking not correct (since it is a straight line and the slope should therefore be the same at all points on the MC curve), but we took this mistake on the graph's values into account during the grading of

the papers and students were awarded marks for any of the three variations of the MC curve's equation.

35.2 Most students were not able to correctly calculate the consumer and producer surplus.

Remember that consumer surplus is defined as the difference between the amount the consumer is willing to pay (as measured through the demand curve), and what the consumer is actually paying (i.e. the market price).

Consumer surplus is therefore the size of the little blue triangle on the graph.

The size of this triangle is calculated as $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 60 \times (200-170) = \mathbf{900}$

Producer surplus is defined as the difference between what producers are willing to accept (as measured through the MC curve, which is also the supply curve) and what they actually receive (which is the market price).

Producer surplus is therefore the sum of the size of the orange triangle and the grey rectangle (that has been coloured in with spots).

The size of the producer surplus area is calculated as $(\frac{1}{2} \times \text{base} \times \text{height}) + \text{the size of the rectangle}$

$$= \frac{1}{2} \times 60 \times (130-10) + [60 \times (170-130)]$$

$$= 3600 + 2400$$

$$= \mathbf{6000}$$

Consumer surplus is therefore = 900

Producer surplus is therefore = 6000

35.3 Most students could not calculate this value correctly. The price mark-up = Price – Marginal cost

$$= 170 - 130$$

$$= 40$$

35.4 Most students got full marks for this question. The deadweight loss is calculated as the sum of the purple triangle and the green triangle.

Deadweight loss = $(\frac{1}{2} \times \text{base} \times \text{height}) + (\frac{1}{2} \times \text{base} \times \text{height})$

$$= [\frac{1}{2} \times (80-60) \times (170-160)] + [\frac{1}{2} \times (80-60) \times (160-130)]$$

$$= 100 + 300$$

$$= 400$$

QUESTION 36

This question was answered very poorly. Many students did not receive any marks, whilst other only received between 1 and 4 marks.

- 36.1 Many students incorrectly said 7 parties, instead of 4.
- 36.2 Oftentimes students said 4 parties, instead of 2.
- 36.3 Students made up stories of how the Jetsams could be benefitted, instead of calculating the decrease in the external cost that is incurred by them. Full marks for an answer of 80 (40+40).
- 36.4 Once again, students gave general suggestions such as having fewer parties, lowering noise levels, agreeing on party times, etc., instead of giving a value between 22 and 80 that could be paid by the Jetsams to the Flotsams to reduce the number of parties from 4 to 2.